

Regression, pt. 2

CSU Chico, Math 351

2026-04-22

outline

Linear Regression Assumptions

Simple Linear Model

References

outline

Linear Regression Assumptions

Simple Linear Model

References

Assumptions

Because we are estimating β_0, β_1 , and σ^2 we know that the estimates are random variables and thus have sampling distributions. In order to draw inference on any of these parameters we need to know the properties of each of the sampling distributions.

Let $(x_1, Y_1), (x_2, Y_2), \dots$, and (x_n, Y_n) be a set of points satisfying the simple linear model, $E(Y|x) = \beta_0 + \beta_1 x$. Let $\hat{\beta}_0, \hat{\beta}_1$, and $\hat{\sigma}^2$ be the maximum likelihood estimators for β_0, β_1 , and σ^2 respectively. Then

1. $\hat{\beta}_0$ and $\hat{\beta}_1$ are both normally distributed.
2. $\hat{\beta}_0, \hat{\beta}_1$ are both unbiased. That is $E(\hat{\beta}_0) = \beta_0$ and $E(\hat{\beta}_1) = \beta_1$.
3. $\text{Var}\hat{\beta}_1 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
4. $\text{Var}\hat{\beta}_0 = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$

outline

Linear Regression Assumptions

Simple Linear Model

References

Simple Linear Model

Theorem 11.3.3: Let $(x_1, Y_1), \dots, (x_n, Y_n)$ satisfy the assumptions of the simple linear model. Then

1. $\hat{\beta}_1$, $\bar{Y}$, and $\hat{\sigma}^2$ are mutually independent.
2. $\frac{n\hat{\sigma}^2}{\sigma^2}$ has a chi square distribution with $n - 2$ degrees of freedom.

Additionally: Let $\hat{\sigma}^2$ be the maximum likelihood estimator for σ^2 in a simple linear model. Then $\frac{n}{n-2}\hat{\sigma}^2$ is an unbiased estimator for σ^2 .

Simple Linear Model

Estimating σ^2 : The unbiased estimator for σ^2 based on $\hat{\sigma}^2$ is denoted as S^2 , where

$$S^2 = \frac{n}{n-2} \hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right)^2$$

Drawing Inference about β_1 : Hypothesis test and confidence intervals for β_1 can be carried out by defining a t statistic based on the previous properties that we learned.

Let $(x_1, Y_1), \dots, (x_n, Y_n)$ be a set of points satisfying the assumptions of the simple linear model, and let

$s^2 = \frac{1}{n-2} \sum_{i=1}^n \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right)^2$. Then

$$T_{n-2} = \frac{\hat{\beta}_1 - \beta_1}{s / \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

has a Student t distribution with $n - 2$ degrees of freedom.

Simple Linear Model

Let $(x_1, Y_1), \dots, (x_n, Y_n)$ be a set of points satisfying the assumptions of the simple linear model, and let

$s^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$. Then

$$\left[\hat{\beta}_1 - t_{\alpha/2, n-2} \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}, \hat{\beta}_1 + t_{\alpha/2, n-2} \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \right]$$

is a $100(1 - \alpha)\%$ confidence interval for β_1 .

outline

Linear Regression Assumptions

Simple Linear Model

References

Exercise 1

Insect flight ability can be measured in a laboratory by attaching the insect to a nearly frictionless rotating arm with a thin wire. The “tethered” insect then flies in circles until exhausted. The nonstop distance flown can easily be calculated from the number of revolutions made by the arm. The following are measurements of this sort made by the arm. The following are measurements of this sort made on *Culex tarsalis* mosquitos of four different ages. The response variable is the average distance flown until exhaustion for forty females of the species.

Age, x (weeks)	Distance Flown, y
1	12.6
2	11.6
3	6.8
4	9.2

Fit a straight line to these data and test that the slope is 0. Use a two-sided alternative and the 0.05 level of significance.

outline

Linear Regression Assumptions

Simple Linear Model

References

references I

Hadley Wickham. *ggplot2: elegant graphics for data analysis*. Springer Science and Business Media, 2009.

David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel. *OpenIntro Statistics*. CreateSpace independent publishing platform, third edition, 2015.

Richard J Larsen, Morris L Marx. *An Introduction to Mathematical Statistics and Its Applications*. Prentice Hall, fifth edition, 2012.

Adapted from notes by Dr. Gray & Dr. Roualdes.